

BrightLearn Business Analytics

Research Assignment 2

Due Date: 26/09/2026

The Language of Business Analytics

Foundational Concepts of Analytics

(01) Business Analytics

Definition- Business Analytics is about examining data, statistics, and numbers with statistical tools and quantitative techniques to help businesses make smarter decisions and perform better.

Real-World example - Retail (Walmart): Analyzes historical point-of-sale data across thousands of stores to determine which products sold best during specific seasons or holidays.

Source: Business Analytics | University of Pretoria | <https://www.up.ac.za>

(02) Data Analytics

Definition: Data Analytics is the practise of looking at raw data to uncover patterns, make meaningful business related conclusions, and help solve business challenges.

NB: Data analytics (DA) is the technical foundation, while Business Intelligence (BI) looks backward to optimize current operations, and Business Analytics looks forward to drive business strategy.

Real-world example - (Netflix and Youtube): These platforms track your viewing habits, search history, and what parts you skip or watch again and they run data through recommendation engines to show you personalized movie or video suggestions next.

Source: Data Analytics | SkillUp | <https://www.skillup.online>

(03) Business Intelligence (BI)

Definition - Business Intelligence (BI) is a process that uses technology to analyze data and provide useful insights that help managers and business leaders make better decisions.

Real-world example - (Target or Walmart) - These companies use BI tools ~~to~~ to track inventory levels, daily sales numbers, and customer purchasing trends across thousands of locations.

Source: Microsoft Power BI/What is Business Intelligence | <https://www.microsoft.com>

(04) Data-Driven Decision Making :- ^{mr}

Definition - Data-Driven Decision Making (DDDM) is the practise of making organizational choices based on real data and facts, not just intuition, observations, or guesswork.

Real-world example - E-commerce Platforms (Amazon) - by analyzing real-time purchasing data, regional demand trends, and browsing history, Amazon's algorithms automatically decide which items to stock in specific fulfilment centers. This data-driven choice ensures fast shipping times and optimizes warehouse space based on actual consumer behavior.

Source: Harvard Business School/The Advantages of Data-Driven Decision-Making | online.hbs.edu

(05) Big Data

Definition - Big Data refers to data that is extremely large, varied, and fast-changing that normal databases and traditional software can't handle effectively.

Real-world example - (Smart Cities and Traffic Management) - They collect massive streams of data from GPS devices in vehicles, public transports smart cards, road sensors, CCTV cameras, and weather feeds simultaneously. High-speed computing platforms ingest this massive variety of live data to adjust traffic light timings in real time, dispatch emergency services instantly, and optimize subway schedules to prevent overcrowding.

Source: Gartner for IT Leaders | [gartner.com](https://www.gartner.com)

(06) Descriptive Analytics

Definition - Descriptive analytics is the process of looking at past and current data to understand what has already happened in a business.

Real-world example - When Netflix launches a new series, executives and product teams need to know exactly how it is performing across the globe. They use descriptive analytics to aggregate millions of raw user interactions into a simple, clear performance report.

Source: What is Descriptive Analytics? | Tableau | <https://www.tableau.com>

(07) Diagnostic Analytics

Definition - Diagnostic analytics is the step after descriptive analytics. Since descriptive analytics tells us what happened, diagnostic analytics explains why it happened by looking deeper into the data to find the root causes, anomalies, and relationships between variables.

Real-world example - There was an instance where an Uber completed rides had dropped by 20% in a specific city on a certain night. This was discovered through Descriptive Analytics. Now the had to discover the reason why it happened

through the use of Diagnostic analytics. The analysts drilled into the data and cross referenced it with other metrics. Then they found that driver cancellations skyrocketed by 300% between 6:00 PM and 8:00 PM. By looking at the external data, they found the root cause: a major downtown storm caused severe traffic delays, making drivers reject long trips to avoid getting stuck.

88. Source: What is Diagnostic analytics | IBM | ibm.com

(08) Predictive Analytics

Definition - Predictive analytics is a specialized field of advanced analytics that uses historical data, statistical modeling, data mining techniques, and machine learning to forecast future outcomes, trends, and behaviors.

Real-world example - A classic, everyday example of predictive analytics is how Netflix predicts what you might want to watch next.

Source: analytics lifecycle | latentview.com

(09) Prescriptive Analytics

Definition - Prescriptive analytics is the most advanced type of analytics. It goes beyond predicting the future, it uses optimization algorithms, simulations, and expert systems to determine what the best action to take is.

Real-world example - Ride-Hailing Apps (Uber/Bolt): When demand spikes in a specific neighborhood, the system uses prescriptive analytics to automatically implement surge pricing and prompt nearby drivers to head toward the high-demand zone.

Source: Gartner for IT Leaders | gartner.com

(10) Data Literacy

Definition - Data literacy is the ability to understand data, to read, write and communicate data in context. It includes understanding data sources, the analytical methods applied, and how to use those findings to drive real-world business value.

Real-world example - A marketing manager looks at a report showing sales dropped by 15% last month, checks that data came from the official sales system, understands what the 15% actually means, and then uses that insight to change the marketing campaign instead of just saying "sales have dropped".

Source: Evaluating Enterprise Data Literacy | Dataversity | <https://www.dataversity.net>

02. Data, Measurement & Reporting

(11) Structured Data

Definition - Structured data is organized data that fits neatly into tables with rows and columns, so it's easy to search and analyze.

Real-world example - Online Shopping Checkouts: When you buy a shirt online, the store's database stores your order as structured data. It has clear fields like Product-ID, Item-Name, Price, Quantity, and Customer-Card-Number.

Source: What is Data Structured vs. Unstructured | Medium | <https://medium.com>

(12) Unstructured Data

Definition - Unstructured data is information that doesn't fit into neat tables or columns, it has no fixed format, so it's messier

to organize.

Real-world example - An email thread: The paragraphs of text, subject lines, and signatures in a conversation with a coworker, that is unstructured data.

Source: What is Unstructured Data? | MongoDB | <https://www.mongodb.com>

(B) Data Warehouse

Definition - A data warehouse is one central place where a company stores large amounts of data from different sources, just to analyze and create reports.

Real-world example - Credit Card Companies; Visa or Mastercard use data warehouses to look at years of your spending history to instantly flag a transaction as "unusual" or fraudulent if it doesn't match your history patterns.

Source: What is a Data Warehouse? | Amazon Web Services (AWS) | <https://aws.amazon.com>

(4) Data Mining

Definition - Data mining is ^{the process} digging through huge amounts of data to find hidden patterns and trends you wouldn't notice just by looking.

Real-world example - Mobile Networks (Vodafone / MTN / Verizon): Mobile operators use data mining to predict customer churn (when a customer is about to cancel their contract and switch to a competitor). The system automatically alerts customer service team to send that specific user a discounted offer before the call to cancel.

Source: What is Data Mining? | Coursera | <https://www.coursera.org>

(15) Data Visualization

Definition - Data visualization refers to turning data into pictures like charts and maps, so it's easier to understand at a glance.

Real-world example - The Weather App: A 7-day temperature line graph that shows you exactly when it will rain or heat up without reading text.

Source: Data Visualization | Coursera | <https://www.coursera.org>

(16) Dashboard

Definition - A dashboard is a visual representation that monitors, analyzes, and displays key performance metrics and data points in real time.

Real-world example - Uber ride app: The live map inside the Uber ride app on a phone. Instead of showing you raw, unreadable GPS coordinates of nearby cars, it takes that complex data and turns it into a simple map with little car icons moving in real time. At a single glance, you instantly see where you are, how close the nearest drivers are, and exactly how many minutes away your ride is.

Source: What is Data Visualization? | IBM | <https://www.ibm.com>

(17) Metric

Definition - A metric is a specific or a standard measurement used to evaluate, quantify and track how well something is performing.

Real-world example - The Business Metrics (Data Analytics): On a powerbi dashboard the metric tells us if the sales of the business are going up or down.

Source: What is a Metric (Data)? | Power Metrics | <https://www.powermetrics.app>

(18) Key Performance Indicator (KPI)

Definition - A Key Performance Indicator (KPI) is a critical, quantifiable metric used to evaluate how well a business or an individual is doing at achieving a core strategic goal.

While every KPI is a metric, not every metric is a KPI.

Real-world example - Monthly New Revenue (MNR) - Imagine you are the Vice President of Sales for a company that sells software to businesses. Your ultimate corporate goal is to grow company wealth. Your primary sales KPI is Monthly New Revenue.

Source: KPIs, OKRs, OMTM: The History of Modern Metrics & Why It Matters | Panoply.io | <https://www.blog.panoply.io>

(19) Benchmark

Definition - A Benchmark is a standard or point of reference used to measure, compare, and judge the quality, value or performance of something.

Real-world example - Inbound to Lead Conversion Rate - The number of leads a Sales team has converted to Opportunities is called a benchmark.

Source: B2B SaaS Conversion Rate Benchmarks 2026 | Orbix Studio | <https://www.>

(20) Segmentation;

Definition - Segmentation refers to dividing a large group of customers into smaller groups that share similar traits

Real-world example - Sales Segmentation; Sales teams segment their targets to assign the right salespeople to the right accounts.

Source: Market segmentation/Wikipedia/<https://www.en.wikipedia.org>

03. Customer & Sales KPIs

(21) Conversion Rate - A conversion rate is the percentage of people who complete the action you desire, like buying or subscribing, out of the total number interactions you had with your leads.

Real-world example - Digital Marketing (E-commerce): when you run a digital ad whether on Instagram or Facebook or whatever social media platform of your choice that directs people to your business website. The total number of people who bought from your website, divided by the total number of people who visited your website will give you the conversion rate of that digital ad.

Source: What is Conversion Rate?/Coursera/<https://www.coursera.org>

(22) Customer Acquisition Cost (CAC)

Definition - A Customer Acquisition Cost is the total amount of money a business spends to gain a new customer.

Real world example - Tyne Bank Strategy: The Hybrid Retail Kiosk Model: Instead of building expensive bank branches, Tyne Bank partnered with

Pay n Pay and Boxer stores. They placed digital, biometric-equipped self-service kiosks directly inside grocery stores where people already do their weekly shopping.

To further assist people who might be hesitant about digital banking, they deployed human "Tyre Bank Ambassadors" right next to the machines.

According to financial reports and statements from executive leadership, Tyre Bank achieves one of the lowest customer acquisition costs not just in South Africa, but globally.

Source: Customer acquisition cost (CAC): calculate and reduce it | Paddle | <https://www.paddle.com>

(23) Customer Lifetime Value (CLV/LTV)

Definition - Customer lifetime value is the total amount a customer is expected to spend with your business over their entire time as a customer in your business.

Real-world example - Starbucks: A classic highly studied real-world example of CAC is Starbucks. When a customer walks into a Starbucks shop and buys a single Coffee Latte for R65, they are not just spending R65, they will most likely come back for another the following day. So the average money a customer spends at Starbucks per visit is R110 (coffee + a snack). The average customer visits a week is 4.2. And a satisfied Starbucks customer stays loyal to the brand for an average of 20 years. So the total revenue a single loyal coffee drinker brings to Starbucks over time is: (Value x Frequency x Lifespan) which is (R110 x 218 visits a year x 20 years = R479 600)

Source: Customer Lifetime Value: formula, calculation & examples | Twilio | <https://www.twilio.com>

(24) Churn Rate

Definition - Churn rate is the percentage of customers who stop using your business or cancel their subscription over a set time. It is the direct opposite of customer retention.

Real-world example - Netflix and Disney+: Historically, Netflix has maintained the lowest churn rate in the entire streaming industry, while its competitors have fought hard to keep users from hitting the cancel button.

- Netflix Monthly churn rate: 2.4%
- Disney+ Monthly churn rate: 4.5%

Source: Netflix Monthly Subscriber Churn Still Lowest at 2% | Yahoo Finance | <https://www.finance.yahoo.com>

(25) Customer Retention Rate

Definition - Customer retention rate is the percentage of customers who stay with your business over a specific period. It is the direct opposite of Churn Rate, while churn rate measures who left, retention rate measures who stayed.

Real-world example - Amazon Prime: According to multi-year consumer data tracking by Consumer Intelligence Research Partners (CIRP), Amazon Prime boasts some of the highest customer retention rates in the entire retail industry.

Source: CIRP - Amazon Report | <https://cirp.amazon.substack.com>

(26) Net Promoter Score (NPS)

Definition - Net Promoter Score is a key metric that measures customer loyalty, satisfaction, and how likely they are to recommend your business to others.

Real-world example - Apple Store: Apple utilizes Net Promoter Score (NPS) to run its retail stores. Apple was one of the earliest large-scale adopters of the system, using it to listen to customers directly after they made a purchase or visited a Genius Bar.

By focusing on making customers happy rather than just pushing products, Apple Retail consistently achieves some of the highest NPS scores in the technology and retail industries, often hovering above 60 or 70.

Source: Net Promoter Score (NPS) & System/Bain | <https://www.bain.com>

(27) Average Order Value

Definition - Average Order Value is a foundational e-commerce metric that tracks the average amount a customer spends each time they place an order.

Real-world example - Pizzeria: At Bella Italia Pizzeria they track their Average Order Value daily by adding up the sum of all the Pizzeria's order on a day and dividing it by the number of pizzas that were ordered ($\frac{\text{Sum/Total Revenue of the day}}{\text{Number of orders made}}$). This gives us the Average Order Value.

Source: Average Order Value: Formula and 7 Ways (2024) | Shopify | <https://www.shopify.com>

28) Sales Funnel

Definition - A Sales Funnel is a step-by-step journey a customer goes through from first discovering your business to making a purchase.

Real-world example - Netflix: Netflix runs one of the most optimized digital sales funnel. Behind the scenes, they track user in four different stages.

- Awareness stage - when a user visits the landing page via a trailer link
- Interest stage - when a user types their email address and clicks "next"
- Desire stage - when a user picks a subscription plan tier.
- Action stage - when a user has finished entering payment info and clicks submit.

This is called a Sales Funnel.

Source: Smart Insights Digital Marketing | <https://www.smartinsights.com>

04. Marketing & Campaign Analytics

(29) Marketing Campaign

Definition - A marketing campaign is a focused, time-bound set of actions designed to promote a specific product, service, or brand goal.

Real-world example - "Share a coke" by The Coca-Cola Company. In year 2011, Coca-Cola came up with an idea of replacing the original Coca-Cola logo with the country's most popular names. This campaign was aimed at reversing a decade-long decline in soda consumption among young adults. And it really worked, people, especially young adults who value personalization and social media posts bought lots of coke just to find one with their name on it and they would then post it on social media with the #shareacoke. This increased sales and soda consumption among young adults like it was intended.

Source: What are Marketing Campaigns? | Salesforce | <https://www.salesforce.com>

(30) Post-Campaign Analysis

Definition- A post-campaign analysis is a review done after a campaign ends to assess its performance, measure return on investment (ROI), and gather insights for future strategies.

Real-world example - A perfect example of a Post-Campaign Analysis is how The Coca-Cola Company evaluated its "Share a Coke" campaign. When the initial campaign wrapped up in its launch market (Australia), the marketing team performed a deep-dive post-campaign analysis. The post-campaign analysis revealed a 7% increase in young adult consumption. Furthermore, overall category share increased by 4%.

Source: Share a Coke campaign post-analysis | Marketing Mag | <https://www.marketingmag.com.au>

(31) Incremental Revenue

Definition- Incremental revenue is the extra income earned as a direct result of a specific campaign or action, on top of what you would have made anyway.

Real-world example - In the context of the "Share a Coke" campaign, incremental revenue is the extra money Coca-Cola made directly because they put names on bottles, excluding the baseline sales they would have made anyway from people who buy Coke regardless. In the United States market, the campaign resulted in an 11% increase in package sales and added more than \$1 billion in total revenue.

Source: Incremental revenue explained: How to measure true marketing impact | SureBright | <https://www.surebright.com>

(32) Return on Investment

Definition - Return on Investment (ROI) is a measure of how profitable an investment was, calculated by dividing the net profit of an investment by its original cost.

In Marketing it answers the ultimate business question: For every Rand or Dollar spent on a campaign, how much money did the business get back.

Real-world - In the context of the Coca-Cola #Share a Coke campaign, independent marketing audits and industry studies confirm that the "Share a Coke" campaign delivered an estimated media ROI of 300% to 400%, translating to a return of over \$4 in incremental revenue for every \$1 spent on the campaign's direct media mix.

Source: Inside Coca-Cola's \$4 Billion Advertising Strategy | Investopedia | <https://www.investopedia.com>

(33) Return on Ad Spend (ROAS)

Definition - Return on Ad Spend (ROAS) measures how much direct revenue you earn for every Rand or Dollar spent on advertising.

Unlike Return on Investment (ROI), which looks at overall profitability of an entire business investment (including manufacturing, staff, and legal fees), ROAS focuses strictly on the efficiency of your direct advertising spend.

Real-world example - When The Coca-Cola Company evaluated the digital and media portions of the "Share a Coke" campaign, ROAS was the primary metric used by the digital marketing teams to see if their paid social media, search, and video ads were working.

As established by independent global marketing audits, the campaign achieved a media ROAS of over 4:1.

Source: How a Groundbreaking Campaign Got Its Start "Down Under"
www.coca-cola.com.au

(34) Click-Through Rate (CTR)

Definition - Click Through Rate is the percentage of people who clicked on an ad, link, or email out of everyone who saw it.

Real-world example - In the context of the Coca-Cola "Share a Coke" campaign, when the campaign was launched on digital ads, CTR was monitored closely to evaluate how well online content drove traffic to their digital asset hubs (such as landing pages where users could digitally customize names or find retail locations stocking specific bottles).

The hyper-targeted homepage placements yielded a 0.85% CTR, outperforming regional video ad click benchmarks by 4x due to the sheer eye-catching relevance of the customized layout.

Source: Teads Omnichannel Boosts 'Share a Coke' in Serbia | Teads | <https://www.teads.com>

(35) Bounce Rate

Definition - Bounce rate is the percentage of visitors who leave a website after viewing just one page, without clicking or engaging further.

Real-world example - Imagine you run an online shoe store called KickZone. A user searches Google for "red running shoes", clicks a link to your store, and lands on your

red running shoes product page. They look at the shoes for 5 seconds, decide they don't like the style, and click the "Back" button on their browser to return to Google. This counts as a bounce.

If there is another user who does the exact something, to calculate the "Bounce Rate" we will determine what percentage of people left the site within a few seconds compared to those who actually bought something from the website.

Source: Bounce rate | Wikipedia | <https://en.wikipedia.org>

(86) A/B Testing

Definition - A/B testing refers to comparing two versions of a webpage, email, or app feature to see which one performs better.

Real-world example - In the context of the red running shoes page, let's imagine the webpage has a high bounce rate. You suspect that users are leaving because the "Add to Cart" button is hard to find. You then run an A/B test to fix it. You then create a second webpage similar to the first one, but on the second one you make the "Add to cart" button brighter in color and bigger, and you make it appear at the top of the page.

You then compare bounce rates on these two webpages, the one with the small "Add to Cart" button at the bottom of the page and the one with the big, bright in color "Add to Cart" button at the top of the page.

This is A/B Testing works in a real-world example.

(37) Lead Generation

Definition - Lead Generation is the process of attracting strangers and turning them into potential customers who have shown interest in your business, usually by sharing their contact details.

Real-world example - A Pizza Shop: A local resident searches Google for "best catering near me" and clicks on your pizza website.

You offer a popup that says, "Planning an event? Enter your email to get a free PDF of our Party Catering Checklist & Pricing."

The user types in their name and email address to get the checklist.

They are no longer a random visitor now, they are a lead because you know they are planning an event and you have a way to contact them.

Source: An Introduction to Lead Generation | Free Guide | HubSpot | <https://www.offers.hubspot.com>

(38) Marketing Attribution

Definition - Marketing attribution is figuring out which marketing channels, ads, or touchpoints actually led a customer to buy or become a lead.

Real-world example - Imagine a customer buys a \$100 pair of shoes from your online store, KickZone. Before buying, they interacted with your brand three times over a week. On Monday they saw an ad on Instagram and clicked it to browse. On Wednesday they read a helpful review on your Blog. And on Friday they searched your brand name on Google, clicked a paid ad, and bought the shoes.

Depending on which Attribution Model your business goes with,

that \$100 sale is credited differently:

- First-Touch Attribution - Instagram gets 100% of the credit. This model values the channel that introduced the customer to your brand.
- Last Touch Attribution - Google Ads gets 100% of the credit. This model values the channel ~~that introduced~~ where the final action that closed the deal was.
- Linear Attribution - Credit is split evenly. Instagram, the Blog, and Google Ads each gets 33.3% of the credit.

Source:  How to Understand Attribution Report Definitions in HubSpot | HubSpot | <https://knowledge.hubspot.com>